

系統程式設計

Lab4 File Attributes

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I. How do you modify fig4_22.c?

Included "sys/stat.h" & "unistd.h" for chmod , access, readlink
added two static long variables valid and invalid and a static long
long variable for total size.

```
added these to print the result
printf("valid\t\t= %ld\n", valid);
printf("invalid\t\t= %ld\n", invalid);
printf("total size\t= %lld\n", total_size);
```

used chmod to change the unreadable dir and tried to reopen the file
chmod(fullpath, S_IRUSR | S_IWUSR | S_IXUSR);

added

```
if (type == FTW_F || type == FTW_D || type == FTW_DNR) { total_size +=
statptr->st_size; }
```

in myfunc to count total size

II. The count of the various types of files?

Ex: File type = Count, Percentage

regular files	=	53, 80.30 %
directories	=	5, 7.58 %
block special	=	0, 0.00 %
char special	=	0, 0.00 %
FIFOs	=	5, 7.58 %
symbolic links	=	3, 4.55 %
sockets	=	0, 0.00 %

```
lk@LK lab4 % make
bash createFile.sh
./fig4_22 `cat /tmp/lk/esllab20140411.sp`
regular files      =      53, 80.30 %
directories         =       5,  7.58 %
block special     =       0,  0.00 %
char special      =       0,  0.00 %
FIFOs             =       5,  7.58 %
symbolic links    =       3,  4.55 %
sockets           =       0,  0.00 %
valid             =       2
invalid           =       1
total size        =    2484
```

III. The count of total files size?

Ex: Total size = Size

Total size = 2484

IV. (Bonus) The count of valid and invalid link?

Ex:

Valid link = Count

Invalid link = Count

Valid link = 2

Invalid link = 1